

AI and Precision Farming have been proposed as solutions to improve agricultural productivity in India. Critically analyse their feasibility and challenges in the Indian agricultural landscape.

As per NITI Aayog, 85% farmers are small and marginal in India. Poor analysis of soil, irrigation pattern are the major reasons for low productivity in India as compared to global level. ~~So~~ Modern technologies like AI and precision farming have the potential to improve agricultural productivity. Precision farming uses digital technology to analyse the crop health, soil health and make agriculture sustainable.

Feasibility of AI and Precision Farming

① Crop monitoring → AI powered drones through real time data and crop health analysis helps in monitoring crops' growth

and identifying issues such as pests, diseases and nutrient deficiencies.

(II) Soil health management → AI driven data would help farmers in understanding soil composition and moisture levels. AI driven soil health management can increase crop yield by 20% and can reduce water & fertiliser usage by 15%.

(III) Efficient irrigation → AI driven data provide information about the amount of water required by specific crop at specific time. As per IWMI, AI based irrigation reduces water wastage upto 30%.

(IV) Pest and Disease control → AI powered cameras can analyse images and detect pests and diseases. For eg → Plantix, an AI powered app detect pest by analysing images.

(V) Reduces post harvest loss → AI can help in determining the perishability period of post harvested crops. This would help in processing food before their wastage.

CHALLENGES FOR AI & PRECISION FARMING

- ① Fragmented land → NITI Aayog reported that 86% farmers are small and marginal. This small holding land prevent them in ~~go~~ acquiring modern technologies.
- ② Initial cost → Small and marginal farmers have lack of institutional credit. Use of Precision farming may put financial burden on them.
- ③ Low internet penetration → Limited internet ~~to~~ connectivity in rural areas is also a major concern for rural farmers.
- ④ Digital illiteracy → Only 14% of rural citizens are active internet users as per PIB.
- ⑤ Red Tapism → Unnecessary regulatory cholesterol act as barrier for farmers to adopt AI.

Thus, proper policy and ~~dego~~ agri digital reforms would help in mitigating these issues. This would help in increasing

yield and would enhance farmer's income,
~~and~~ will reduce environment impacts and
thus would make agricultural sector more
sustainable.

"Water Scarcity is the one of the major challenges in Indian agriculture. Evaluate the role of micro-irrigation techniques in addressing this issue and discuss the policy measures needed for their widespread adoption."

India has population of 1.4 billion people but has only 4% of the world's freshwater resource. As India is net exporter of water intensive crops, so this export of virtual water is also a major reason for groundwater depletion of Indian water resource.

To address this issue, specific crops need specific amount of water through technique of micro-irrigation.

Role of Micro-irrigation technique

① Water efficient → Use of micro-irrigation like drip irrigation system increases water efficiency by 50-60%.

⑪ Water saving → As

- As per Ashoka Dalwai committee report, use of microirrigation can save upto 70% of water compared to traditional flood irrigation.
- Use of sprinkler system could decrease water wastage by 30-40%.

⑫ Improves fertigation system → ~~Use of~~

Integrating fertiliser with drip irrigation system to spread fertiliser directly to plant's root helps in utilising the optimal amount of fertiliser.

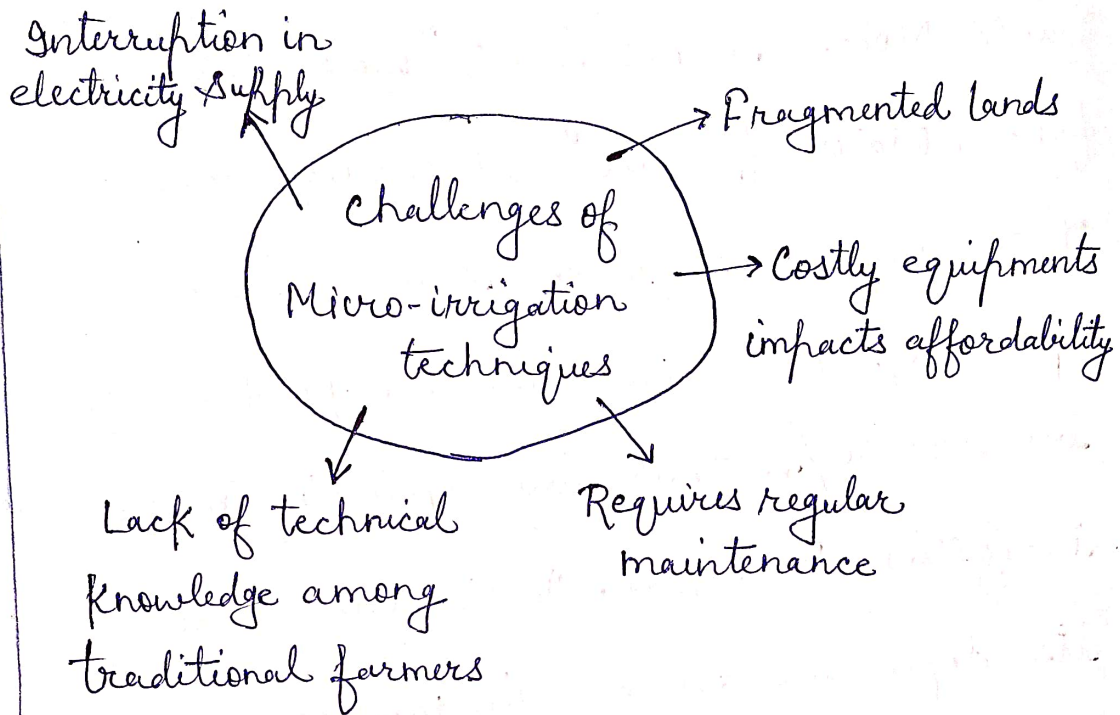
⑬ Energy efficient → Microirrigation

reduces ~~electro~~ energy use and thus electricity cost for farmers.

⑭ Crop productivity → Micro irrigation

~~to~~ enhances crop productivity by 40%.

- (vi) Increases farmer's income → optimal use of fertiliser and cost effective electricity by micro-irrigation technique could increase farmer's income.



POLICY MEASURES

- (i) Increase financial support for 'Per Drop More Crop scheme' under PMKSY, ensuring 80% subsidy for small and marginal farmers.
- (ii) Introduce custom hiring centres, so that small and marginal farmers could have access to micro-irrigation tools at low rental price.

③ Government of India should enhance grant on loans, which is availed by states from NABARD's Micro Irrigation Fund (MIF).

David Molden in his book "Water for food, Water for life" has suggested that smart irrigation can save water upto 50% and can increase productivity by 25-30%. Thus, by expanding micro-irrigation techniques, India can reduce its dependence on erratic monsoons and make its agriculture water efficient.